TARUN BAYYA
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PROFESSIONAL SUMMARY

Mechanical Engineer with 5+ years of industry experience delivering manufacturing-ready design for industrial systems, including gaseous hydrogen generation. Experienced in end-to-end engineering documentation from PFD/P&ID development through interlocks and controls documentation, translating process intent into buildable, safe, and serviceable designs. Brings a working understanding of semiconductor manufacturing principles and process control in precision production environments. Delivered a 15% cutting-cost reduction through manufacturability-driven design changes and a 16% error reduction through disciplined drawing execution and cross-functional coordination.

TECHNICAL SKILLS

Mechanical Tools: AutoCAD, ACAD-E, Plant 3D, SolidWorks, SolidWorks PDM, Siemens NX, Teamcenter, CATIA V5/V6.

CAE/Simulation: ANSYS Mechanical, Altair HyperWorks, Siemens Plant Simulation.

Data / Automation / Analytics: Python, SQL/MySQL, Tableau, Power BI, Minitab, R Studio, Apache Airflow, Google NLP.

Manufacturing / Quality Methods: DFMA, DFM/DFA, GD&T, DFMEA/PFMEA, DMAIC, Root Cause Analysis, Value Stream Mapping, Wet Etching, SPC, FDC, Statistical Analysis, SOP Development, Material Selection, ISO 9001 & 14001.

PROFESSIONAL EXPERIENCE

OneH2, Inc. **Clover, SC, USA**
Mechanical Engineer - Manufacturing & Engineering Design *August 2024 - February 2026*

- Developed and maintained **PFDs and P&IDs 2D CAD** documentation for **gaseous hydrogen generation** systems, translating process into instrumented drawings aligned with **SAE, ASME, NEC, NFPA & OSHA** requirements.
- Created and updated **electrical interlocks** using AutoCAD Electrical & authored the **H600 Control Narrative** and **I/O table** to clarify signal intent for embedded engineers while code implementation.
- Built and optimized **3D CAD assemblies in SolidWorks,** including **sheet metal design** considerations, by improving **maintenance access, safety clearances, and component serviceability** across the skid layout.
- Supported site commissioning by conducting **HAZOP** test and developing **start-up and shut-down process documentation** to strengthen operational safety and readiness.
- Applied **DFMEA** during design updates to identify failure modes and define practical controls that improved build consistency & reduced assembly issues.
- Collaborated with electrical engineers on **circuit board placement** to improve **wiring accessibility** & assembly feasibility.
- Applied **Dfx principles**, including **DFM & DFA, GD&T - based tolerance control** to improve manufacturability and scalability, reducing **cutting cost by 15%** while improving assembly repeatability.
- Collaborated with **R&D** to perform **CFD and FEA simulations in ANSYS Mechanical** to **analyze flow, pressure loss**, and **heat dissipation**, and specified heat exchanger insulation to maintain performance across varying climatic conditions.
- Improved **2D/3D CAD** drawing quality and on time release execution across **AutoCAD Electrical and SolidWorks** in an **ISO 9001 - aligned environment**, reducing errors by **16%** and improving cross-team communication.

Clemson University **Clemson, SC, USA**
Graduate Research Assistant *August 2022 - December 2023*

- Conducted literature review on **Non - Functional Requirements (NFRs)** in automotive design and presented "**An Overview on ilities**" at **VIPR - GS**, communicating complex findings for a mixed audience.
- Reviewed **Deep Orange** conceptual vehicle documents to extract "ilities" phrases; categorized them in **Excel** and resolved differences through a **two - reviewer** comparison workflow.
- Performed **text mining with Google NLP** and built a **Python - based ML model** to automate identification & classification of "ilities" and NFR - style phrases, improving consistency and throughput.
- Developed a **KAOS goal - model** case study, mapping ilities to software modules, executed **DOE** across **three ocean - current environments** and used **R Studio Kruskal - Wallis** test to compare performance distributions.

Amazon **Hyderabad, TG, India**
Operations Engineer *June 2021 - July 2022*

- Managed SPS - Seller Performance Support operations, resolving customer complaints and product issues while protecting marketplace compliance and customer trust.
- Used **SIM ticketing tools** to track, analyze, and quality mismatch issues with structured documentation and follow up.
- Performed root cause analysis and streamlined escalations and partnered with cross - functional teams to implement corrective actions, improving resolution efficiency and reducing repeat complaints.
- Applied **DMAIC** to supply chain logistics, **reducing defects and increasing efficiency by 22%** through standardization.

SRK Constructions & Projects Pvt. Ltd. **Chandrapur, MH, India**
Project Design Engineer *July 2020 - April 2021*

- Engineered underground pipeline projects in remote terrains, integrating **MEMS sensors, PLC, and SCADA** for real - time monitoring and decision support.
- Built **PERT Chart based schedules** and improved project execution using **SAP ERP** for **materials management, inventory & procurement tracking, and resource planning**, supporting on - time delivery.
- Delivered **12% improvement ($2.3M savings)** by redesigning pipeline installation processes and applying lean practices.
- Led cross - functional coordination and stakeholder engagement to secure approvals and minimize environmental impact.

LANTEK Engineering Consultants **Hyderabad, TG, India**
Mechanical Design Engineer *April 2019 - May 2020*

- Led **IQ, OQ & PQ** validations, ensuring **product compliance** and driving cost - saving design improvements.
- Enhanced **tooling durability** through **data - driven analysis**, cutting downtime & **increasing** fixture **lifespan by 20%.**
- **Redesigned fixtures**, optimizing geometries to eliminate **bottlenecks**, improving **production throughput** by **18%,** and reducing cycle time.
- Validated manufacturability for **New Product Introduction (NPI)**, aligning designs with production for smooth scale - up.

Null Innovation Inc. **Hyderabad, TG, India**
Data Analyst *November 2018 - March 2019*

- Automated stock price extraction using **Apache Airflow**, improving repeatability and reducing manual effort in data pulls.
- Built pipelines using **Python, SQL, Tableau, Google Analytics, and R Studio** to support extraction, transformation, and reporting workflows.

RESEARCH & ACADEMIC PROJECTS

Comprehensive Structural & thermal analysis of the Body-in-White of a 2016 Honda Civic using Siemens NX

- Personally executed **experimental BIW torsion testing** using **LVDTs**, calculated twist angles, and extracted torsional stiffness from torque - twist curves.
- Conducted multi - step bending tests (up to **978 N**) and validated linear response and repeatability across trials to ensure reliable stiffness results.
- Developed and evaluated subsystem weight - saving proposals and calculated a **61.5 kg net mass reduction** after accounting for required component weight increases.

Optimization of Bell Housing using Siemens Plant Simulation software

- Built a **discrete event simulation** of bell housing machining and assembly to meet **70,000 units/year**, comparing **ideal, typical, and cost - optimized** operating scenarios.
- Improved throughput and cost performance by balancing the line and optimizing equipment & labor, achieving daily output targets while delivering significant capital and monthly operating cost savings.

PLM Change Management Project (RYOBI Handled Sprayer) using Siemens Teamcenter

- Built a **PLM documentation baseline** for the RYOBI ONE+ 18V handheld sprayer, including **BOM** and **part identification** and traceable **requirements (REQ tagging)** to define system functionality and constraints.
- Executed an **engineering change (ECO - 00693)** to improve trigger ergonomics and validated the update using **NX assembly checks** and **3D-printed prototypes**, supported by tolerance/cost verification and Python-based reliability analysis.

Effect of Pulsed Micro Plasma Arc Parameters on Weld Bead Geometry of AISI 316Ti Austenitic Stainless Steel

- Co-authored a peer-reviewed study on **0.3 mm AISI 316Ti**, executing a **27-run Box-Behnken DOE** and measuring weld bead geometry. Built prediction models in **MINITAB** and validated results using **ANOVA (95% confidence)** to define an optimal operating window.

EDUCATION

Clemson University **August 2022 - December 2023**
Master's, Mechanical Engineering *GPA: 3.42/4.0*

Andhra University **August 2013 - November 2017**
Bachelor's, Mechanical Engineering *CGPA: 6.49/8.0*

CERTIFICATIONS

- Six Sigma Green Belt Certification
- Supply Chain Logistics in R/3
- Master Certification in Computer Aided Tool Engineering
- Lean Manufacturing
- Supply Chain Analytics using Excel & Python
- Data Science Pro-Degree Certification